

What Makes Training Multi-modal Classification Networks Hard?

W. Wang, D. Tran, and M. Feiszli, CVPR, 2020

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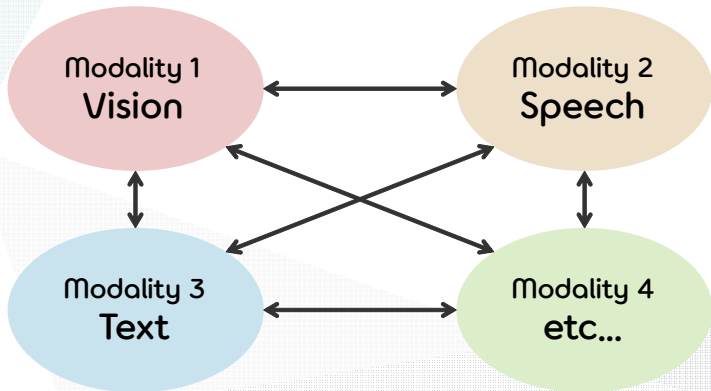
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1. Introduction: what is multi-modal?



1. Introduction

- ▶ Multi-modal not always outperform the best uni-modal model
 - Overfitting problem
 - multi-modal networks have higher train accuracy and lower validation accuracy
 - Solution
 - Dropout, pre-training, or early stopping
 - Approach in terms of structural defects
 - Mid-level fusion by concatenation and fusion by gating, trying both Squeeze-and-Excitation (SE) gates and Non-Local (NL) gates

Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
	RGB + OF	71.3	RGB	72.6	-1.3
	A + OF	58.3	OF	62.1	-3.8
	A + RGB + OF	70.0	RGB	72.6	-2.6

Table 1: Uni-modal networks consistently outperform multimodal networks. Best uni-modal networks vs late fusion multimodal networks on Kinetics using video top-1 validation accuracy. Single stream modalities include video clips (RGB), Optical Flow (OF), and Audio (A). Multi-modal networks use the same architectures as uni-modal, with late fusion by concatenation at the last layer before prediction.

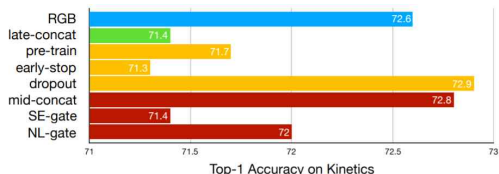


Figure 1: Standard regularizers do not provide a good improvement over the best Uni-modal network. Best uni-modal network (RGB) vs standard approaches on a multi-modal network (RGB+Audio) on Kinetics. Various methods to avoid overfitting (orange: Pre-training, Early-stopping, and Dropout) do not solve the issue. Different fusion architectures (red: Midconcatenation fusion, SE-gate, and NL-gate) also do not help. Dropout and Mid-concatenation fusion approaches provide small improvements (+0.3% and +0.2%), while other methods degrade accuracy.

1. Introduction: Contribution

- ▶ Empirically demonstrate the significance of **overfitting in joint training of multi-modal networks**, and we identify two causes for the problem
- ▶ Propose a metric to understand the problem quantitatively: the **overfitting-to-generalization ratio (OGR)**
- ▶ Propose a **new training scheme which minimizes OGR** via an optimal blend of multiple supervision signals
- ▶ Proposed Gradient-Blending (G-Blend) method gives significant gains in ablations and achieves state-of-the-art (SoTA) accuracy on benchmarks including Kinetics, EPIC-Kitchen, and AudioSet by combining audio and visual signals.

2. Multi-modal training via Gradient-Blending

► Background

■ Uni-modal network

$$T = \{X_{1,...,n}, Y_{1,...,n}\}$$

T : train set / X_i : i th training example / Y_i : true label

$$\mathcal{L}(\mathcal{C}(\varphi_m(X)), y) \quad (1)$$

m : single modality / φ_m : deep network with parameter θ_m

$\mathcal{C}$: classifier, typically one or more FC layers with parameter θ_c

$\mathcal{L}$: cross entropy loss

■ Multi-modal network (late-fusion model)

$$\mathcal{L}_{multi} = \mathcal{L}(\mathcal{C}(\varphi_{m_1} \oplus \varphi_{m_2} \oplus \dots \oplus \varphi_{m_k}), y) \quad (2)$$

$\{m_i\}_1^k$: M different modalities / φ_m : deep network with parameter θ_m

$\mathcal{C}$: classifier / $\oplus$: a fusion operation (e.g. concatenation)

- The multi-modal network in Eq. 2 is a super-set of the uni-modal network in Eq. 1.
- for any solution to Eq. 1 on any modality m_i , one can construct an equally-good solution to Eq. 2 by choosing parameters θ_c that mute all modalities other than m_i .
- In practice, this solution is not found, and we next explain why.

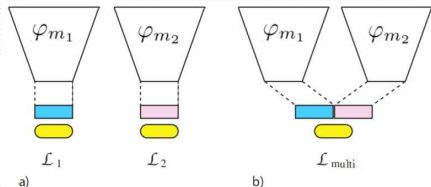


Figure 2: Uni- vs. multi-modal joint training. a) Uni-modal training of two different modalities. b) Naive joint training of two modalities by late fusion. Different deep network encoders (white trapezoids) produce features (blue or pink rectangles) which are concatenated and passed to a classifier (yellow rounded rectangles).

2. Multi-modal training via Gradient-Blending

► Generalizing vs. Overfitting

- Overfitting: learning patterns in a train set that do not generalize to the target distribution

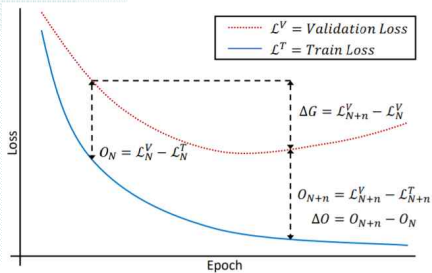


Figure 3: Overfitting-to-Generalization Ratio. Between any two training checkpoints, we can measure the change in overfitting and generalization. When $\Delta O / \Delta V$ is small, the network is learning well and not overfitting much.

► Overfitting-to-generalization-ratio (OGR):

$$OGR \equiv \left| \frac{\Delta O_{N,n}}{\Delta G_{N,n}} \right| = \left| \frac{O_{N+n} - O_N}{\mathcal{L}_N^* - \mathcal{L}_{N+n}^*} \right| \quad (3)$$

$$OGR^2 = \left(\frac{\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, \hat{g} \rangle}{\langle \nabla \mathcal{L}^*, \hat{g} \rangle} \right)^2 \quad (4)$$

2. Multi-modal training via Gradient-Blending

► Blending of Multiple Supervision Signals by OGR Minimization

Proposition 1 (Optimal Gradient Blend). *Let $\{v_k\}_0^M$ be a set of estimates for $\nabla \mathcal{L}^*$ whose overfitting satisfies $\mathbb{E}[\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_k \rangle \langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_j \rangle] = 0$ for $j \neq k$. Given the constraint $\sum_k w_k = 1$ the optimal weights $w_k \in \mathbb{R}$ for the problem*

$$w^* = \arg \min_w \mathbb{E} \left[\left(\frac{\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, \sum_k w_k v_k \rangle}{\langle \nabla \mathcal{L}^*, \sum_k w_k v_k \rangle} \right)^2 \right] \quad (5)$$

are given by

$$w_k^* = \frac{1}{Z} \frac{\langle \nabla \mathcal{L}^*, v_k \rangle}{\sigma_k^2} \quad (6)$$

where $\sigma_k^2 \equiv \mathbb{E}[\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_k \rangle^2]$ and $Z = \sum_k \frac{\langle \nabla \mathcal{L}^*, v_k \rangle}{2\sigma_k^2}$ is a normalizing constant.

Assumption $\mathbb{E}[\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_k \rangle \langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_j \rangle] = 0$ will be false when two models' overfitting is very correlated. However, if this is the case then very little can be gained by blending their gradients. In informal experiments we have indeed observed that these cross terms are often small relative to the $\mathbb{E}[\langle \nabla \mathcal{L}^T - \nabla \mathcal{L}^*, v_k \rangle^2]$.

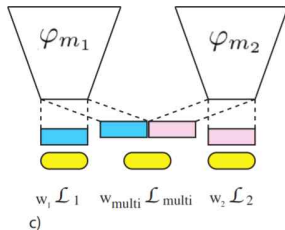


Figure 2: Multi-modal joint training. c) Joint training of two modalities with weighted blending of supervision signals. Different deep network encoders (white trapezoids) produce features (blue or pink rectangles) which are concatenated and passed to a classifier (yellow rounded rectangles).

2. Multi-modal training via Gradient-Blending

- Use of OGR and Gradient-Blending in practice
 - Optimal blending by loss re-weighting

Algorithm 1: G-B Weight Estimation: *GB_Estimate*

input: φ^N , Model checkpoint at epoch N
 n , # of epochs

Result: A set of optimal weights with for $k + 1$ losses.

for $i = 1, \dots, k + 1$ **do**

 Initialize uni-modal/ naive multi-modal network

$\varphi_{m_i}^N$ from corresponding parameters in φ^N ;

 Train $\varphi_{m_i}^N$ for n epochs on $\mathcal{T}$, resulting model

$\varphi_{m_i}^{N+n}$;

 Compute amount of overfitting $O^i = O_{N,n}$,
 generalization $G^i = G_{N,n}$ according to Eq.3
 using $\mathcal{V}$ and $\mathcal{T}'$ for modality m_i ;

end

 Compute a set of loss $\{w_i^*\}_{i=1}^{k+1} = \frac{1}{Z} \frac{G^i}{O^i{}^2}$;

$$\mathcal{L}_{blend} = \sum_{i=1}^{k+1} w_i \mathcal{L}_i \quad (7)$$

2. Multi-modal training via Gradient-Blending

► Use of OGR and Gradient-Blending in practice

- Measuring OGR in practice
 - Offline Gradient-Blending: a simple version of gradient-blending. We compute weights only once, and use a fixed set of weights to train entire epoch (Very easy to implement).
 - Online Gradient-Blending: the full version. We re-compute weights regularly (e.g. every n epochs called a super-epoch), and train the model with new weights for a super-epoch.

Algorithm 2: Offline Gradient-Blending

input: φ^0 , Initialized model
 N , # of epochs

Result: Trained multi-head model φ^N

Compute per-modality weights

$\{w_i\}_{i=1}^k = GB_Estimate(\varphi^0, N)$;

Train φ^0 with $\{w_i\}_{i=1}^k$ for N epochs to get φ^N ;

Algorithm 3: Online Gradient-Blending

input: φ^0 , Initialized model
 N , # of epochs
 n , super-epoch length

for $i = 0, \dots, \frac{N}{n} - 1$ **do**

 Current epoch $N_i = i * n$;

 Compute per-modality weights

$\{w_i\}_{i=1}^k = GB_Estimate(\varphi^{N_i}, N_i + n)$;

 Train φ^{N_i} with $\{w_i\}_{i=1}^k$ for n epochs to φ^{N_i+n} ;

end

3. Ablation Experiments

► Experimental setup

- Datasets. Kinetics, mini-Sports, and mini-AudioSet
 - **Kinetics**: action recognition with 260k videos of 400 human action classes (training and validation: 240k, testing: 20k)
 - **Mini-Sports**: (a subset of Sports-1M) a large-scale classification dataset with 1.1M videos of 487 different finegrained sports (training 240k videos, test: 20k videos)
 - **Mini-AudioSet**: (a subset of AudioSet) 2M videos labeled by 527 acoustic events (class-unbalanced, so removed tiny classes and subsample the rest) → 418 classes with 243 videos

Kinetics example (riding a bike) [2]



Mini-Sports example [3]



3. Ablation Experiments (cont.)

► Experimental setup (cont.)

- Input preprocessing & augmentation
 - RGB, optical flow and audio
 - Input clip size of RGB and optical flow: 16 x 224 x 224
 - Audio: log-Mel with 100 temporal frames by 40 Mel filters
 - Audio and visual are temporally aligned
- Backbone architecture
 - ResNet3D (50 layers): visual backbone for RGB and flow
 - ResNet (50 layers): audio model
 - Fusion: two FC-layer network on concatenated features from visual and audio backbones, followed by one prediction layer
- Training and testing
 - We train our models with synchronous distributed SGD on GPU clusters using Caffe2. We hold out a small portion of training data for weight estimate (8% for Kinetics and mini-Sports, 13% for mini-AudioSet). The final video prediction is made by using center crops of 10 uniformly sampled clips and averaging the 10 predictions.

3. Ablation Experiments (cont.)

- Overfitting Problems in Naive Joint Training

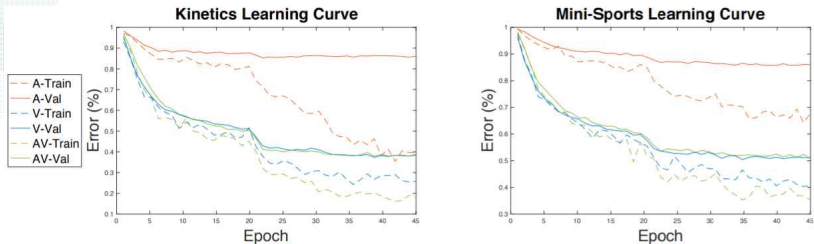


Figure 4: Severe overfitting of naive audio-video models on Kinetics and mini-Sports. The learning curves (error-rate) of audio model (A), video model (V), and the naive joint audio-video (AV) model on Kinetics (left) and mini-Sports (right). Solid lines plot validation error while dashed lines show train error. The audio-video model overfits more than visual model and is inferior to the video-only model on validation loss.

3. Ablation Experiments (cont.)

- Overfitting Problems in Naive Joint Training

Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
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3. Ablation Experiments (cont.)

- Gradient-Blending is an effective regularizer
 - Online G-Blend Works.

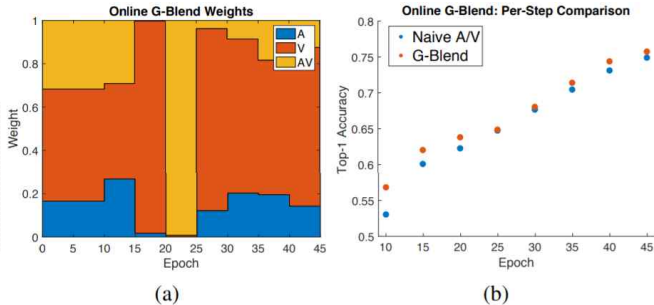


Figure 5: Online G-Blend. (a) Online G-Blend weights for each head. (b) Online G-Blend outperforms naive training on each super-epoch. For each super-epoch (5 epochs), we use the same snapshot of the model learned by G-Blend, and compare the performance of the models trained by G-Blend and naive at the next 5 epochs. G-Blend always outperforms naive training. This proves that G-Blend always learn more generalize information at a per-step level.

3. Ablation Experiments (cont.)

- Gradient-Blending is an effective regularizer
 - Offline G-Blend Also Works.

Optimizer	Method	Clip	V@1	V@5
AdaGrad	Visual	60.0	68.9	88.4
	Naive AV	56.4	65.2	86.5
	G-Blend	62.1	71.3	89.8
Adam	Visual	60.1	69.3	88.7
	Naive AV	57.9	66.4	86.8
	G-Blend	63.0	72.1	90.5

Table 2: Both offline and online Gradient-Blending outperform Naive late fusion and RGB only. Offline G-Blend is lightly less accurate compared with the online version, but much simpler to implement.

- Adaptive Optimizers.

3. Ablation Experiments (cont.)

► Gradient-Blending is an effective regularizer

- Different Modalities.

Modal	RGB + A			RGB + OF			OF + A			RGB + OF + A		
Weights	[RGB,A,Join]=[0.630,0.014,0.356]			[RGB,OF,Join]=[0.309,0.495,0.196]			[OF,A,Join]=[0.827,0.011,0.162]			[RGB,OF,A,Join]=[0.33,0.53,0.01,0.13]		
Metric	Clip	V@1	V@5	Clip	V@1	V@5	Clip	V@1	V@5	Clip	V@1	V@5
Uni	63.5	72.6	90.1	63.5	72.6	90.1	49.2	62.1	82.6	63.5	72.6	90.1
Naive	61.8	71.4	89.3	62.2	71.3	89.6	46.2	58.3	79.9	61.0	70.0	88.7
G-Blend	65.9	74.7	91.5	64.3	73.1	90.8	54.4	66.3	86.0	66.1	74.9	91.8

Table 4: Gradient-Blending (G-Blend) works on different multi-modal problems. Comparison between G-Blend with naive late fusion and single best modality on Kinetics. On all 4 combinations of different modalities, G-Blend outperforms both naive late fusion network and best uni-modal network by large margins, and it also works for cases with more than two modalities. G-Blend results are averaged over three runs with different initialization.

- Different Architectures.

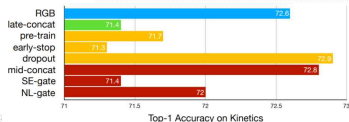


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3. Ablation Experiments (cont.)

- Gradient-Blending is an effective regularizer
 - Different Tasks/Benchmarks.

Dataset	Kinetics			mini-Sports			mini-AudioSet	
Weights	[RGB,A,Join]=[0.63,0.01,0.36]			[RGB,A,Join]=[0.65,0.06,0.29]			[RGB,A,Join]=[0.38,0.24,0.38]	
Method	Clip	V@1	V@5	Clip	V@1	V@5	mAP	mAUC
Audio only	13.9	19.7	33.6	14.7	22.1	35.6	29.1	90.4
RGB only	63.5	72.6	90.1	48.5	62.7	84.8	22.1	86.1
Pre-Training	61.9	71.7	89.6	48.3	61.3	84.9	37.4	91.7
Naive	61.8	71.7	89.3	47.1	60.2	83.3	36.5	92.2
Dropout	63.8	72.9	90.6	47.4	61.4	84.3	36.7	92.3
Auxiliary Loss	60.5	70.8	88.6	48.9	62.1	84.0	37.7	92.3
G-Blend	65.9	74.7	91.5	49.7	62.8	85.5	37.8	92.2

Table 5: G-Blend outperforms all baseline methods on different benchmarks and tasks. Comparison of G-blend with different regularization baselines as well as uni-modal networks on Kinetics, mini-Sports, and mini-AudioSet. G-Blend consistently outperforms other methods, except for being comparable with using auxiliary loss on mini-AudioSet due to the similarity of learned weights of G-Blend and equal weights.

3. Ablation Experiments (cont.)

- Gradient-Blending is an effective regularizer
 - Different Tasks/Benchmarks.

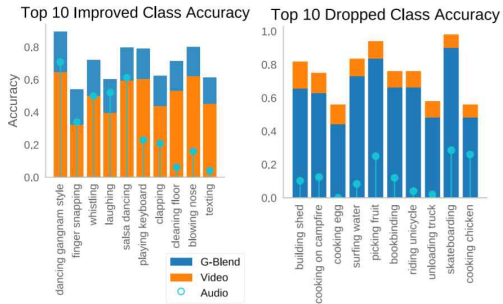


Figure 6: Top-Bottom 10 classes based on improvement of G-Blend to RGB model. The improved classes are indeed audio-relevant, while those have performance drop are not very audio semantically-related.

4. Comparison with State-of-the-Art

Backbone	Pre-train	V@1	V@5	GFLOPs
Shift-Attn Net [10]	ImageNet	77.7	93.2	NA
SlowFast [17]	None	78.9	93.5	213×30
SlowFast+NL [17]	None	79.8	93.9	234×30
ip-CSN-152 [46]	None	77.8	92.8	108.8×30
G-Blend(ours)	None	79.1	93.9	110.1×30
ip-CSN-152 [46]	Sports1M	79.2	93.8	108.8×30
G-Blend(ours)	Sports1M	80.4	94.8	110.1×30
ip-CSN-152 [46]	IG-65M	82.5	95.3	108.8×30
G-Blend(ours)	IG-65M	83.3	96.0	110.1×30

Table 6: Comparison with state-of-the-art methods on Kinetics. GBlend used audio and RGB as input modalities; for pre-trained models on Sports1M and IG-65M, G-Blend initializes audio network by pre-training on AudioSet. G-Blend outperforms current state-of-the-art multi-modal method (Shift-Attention Network) despite the fact that it uses fewer modalities (G-Blend does not use Optical Flow). G-Blend also gives a good improvement over RGB model (the best uni-modal network) when using the same backbone, and it achieves the state-of-the-arts.

Method	mAP	mAUC
Multi-level Attn. [55]	0.360	0.970
TAL-Net [52]	0.362	0.965
Audio:R2D-101	0.324	0.961
Visual:R(2+1)D-101	0.188	0.918
Naive A/V:101	0.402	0.973
G-Blend (ours):101	0.418	0.975

Table 7: Comparison with state-of-the-art methods on AudioSet. GBlend outperforms the state-of-the-art methods by a large margin.

4. Comparison with State-of-the-Art

method	noun		verb		action	
	V@1	V@5	V@1	V@5	V@1	V@5
Validation Set						
Visual:ip-CSN-152 [46]	36.4	58.9	56.6	84.1	24.9	42.5
Naive A/V:152	34.8	56.7	57.4	83.3	23.7	41.2
G-Blend(ours)	<u>36.1</u>	<u>58.5</u>	59.2	84.5	25.6	43.5
Test Unseen Kitchen (S2)						
Leaderboard [2]	38.1	63.8	60.0	<u>82.0</u>	27.4	<u>45.2</u>
Baidu-UTS [51]	34.1	<u>62.4</u>	59.7	82.7	25.1	46.0
TBN Single [29]	27.9	53.8	52.7	79.9	19.1	36.5
TBN Ensemble [29]	30.4	55.7	54.5	81.2	21.0	39.4
Visual:ip-CSN-152	35.8	59.6	56.2	80.9	25.1	41.2
G-Blend(ours)	<u>36.7</u>	60.3	58.3	81.3	<u>26.6</u>	43.6
Test Seen Kitchen (S1)						
Baidu-UTS(leaderboard)	52.3	76.7	69.8	91.0	41.4	63.6
TBN Single	46.0	71.3	64.8	90.7	34.8	56.7
TBN Ensemble	47.9	<u>72.8</u>	66.1	91.2	36.7	<u>58.6</u>
Visual:ip-CSN-152	45.1	68.4	64.5	88.1	34.4	52.7
G-Blend(ours)	<u>48.5</u>	71.4	<u>66.7</u>	88.9	<u>37.1</u>	56.2

Table 8: Comparison with state-of-the-art methods on EPIC-Kitchen. G-Blend achieves 2nd place on seen kitchen challenge and 4th place on unseen, despite using fewer modalities, fewer backbones, and single model in contrast to model ensembles compared to published results on leaderboard

References

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- [5] Tran, Du, et al. "Video classification with channel-separated convolutional networks." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2019.

Supplementary materials

► Visual pre-processing and augmentation [5]:

- We use both spatial and temporal jittering for augmentation. Specifically, video frames are scaled such that the shorter edge of the frames becomes s while we maintain the frame original aspect ratio. During training, s is uniformly sampled between 256 and 320. Each clip is then generated by randomly cropping windows of size 224×224 . Temporal jittering is also applied during training by randomly selecting a starting frame and decoding T frames. For the ablation experiments in this section we train and evaluate models with clips of 8 frames ($T = 8$) by skipping every other frame (all videos are pre-processed to 30fps, so the newly-formed clips are effectively at 15fps).